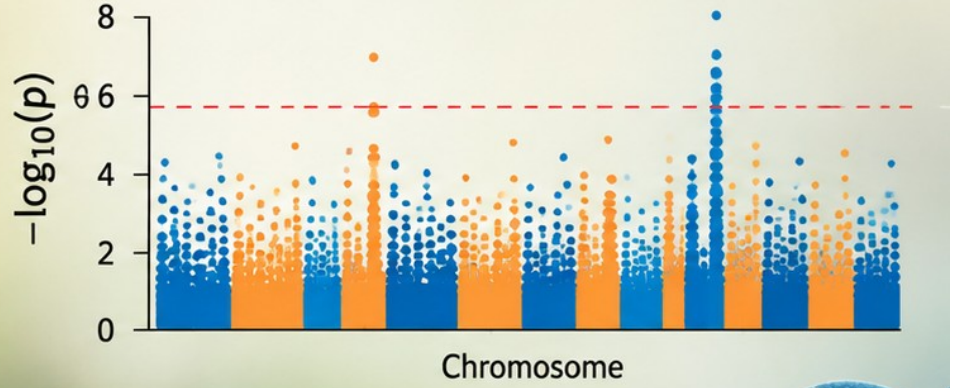
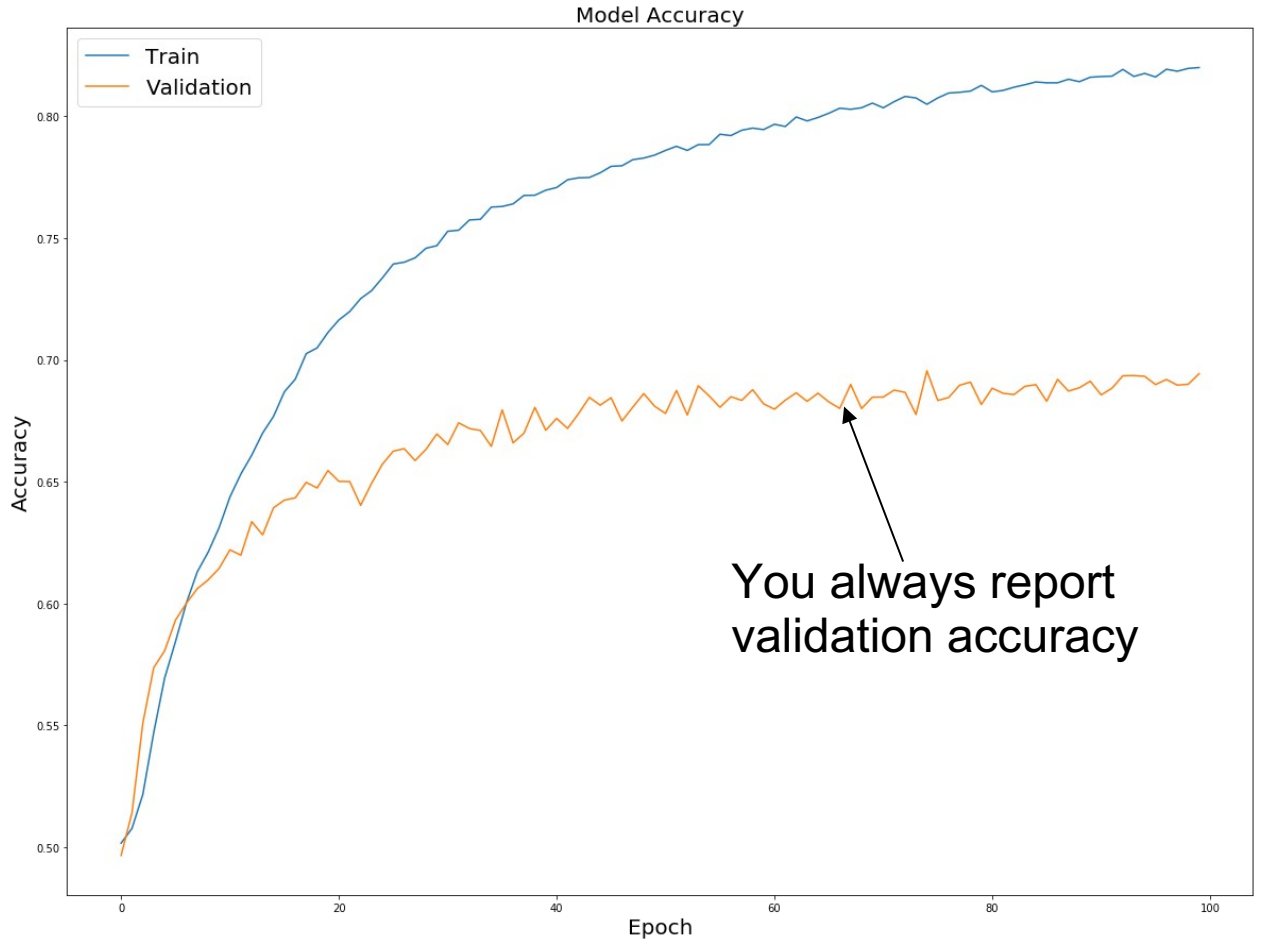
# Genome-Wide Association Studies (GWAS) for Evolutionary Biology

Nikolay Oskolkov, Group Leader (PI) at NIRI, Riga, Latvia





# Accuracy of prediction in machine learning







Explained variance of phenotype, **y**, has something to do with prediction: $R^2 = 1 - (y - (a + b*x))^2 / (y - E(y))^2$

## Principal Investigator




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We found three genetic variants that explain 18%, 12% and 15% of phenotypic variation in herring

Genetic variants associated with human traits explain negligible %, this is different for non-model organisms

Leif Andersson


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The variant explains 20% of variation

On an independent sample, right?





On an independent sample, right?

nature genetics

# Defining the role of common variation in the genomic and biological architecture of adult human height

**Using genome-wide data from 253,288 individuals, we identified 697 variants at genome-wide significance that together explained one-fifth of the heritability for adult height. By testing different numbers of variants in independent studies, we show that the most strongly associated ~2,000, ~3,700 and ~9,500 SNPs explained ~21%, ~24% and ~29% of phenotypic variance. Furthermore, all common variants together captured 60% of heritability. The 697 variants clustered in 423 loci were enriched for genes, pathways and tissue types known to be involved in growth and together implicated genes and pathways not highlighted in earlier efforts, such as signaling by fibroblast growth factors, WNT/β-catenin and chondroitin sulfate–related genes. We identified several genes and pathways not previously connected with human skeletal growth, including mTOR, osteoglycin and binding of hyaluronic acid. Our results indicate a genetic architecture for human height that is characterized by a very large but finite number (thousands) of causal variants.**

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Height is a classical polygenic trait that has provided general insights into the genetic architecture of common human traits and diseases and into the prospects and challenges of different methods used to identify genetic risk factors. Studies consistently estimate that the additive genetic contribution to normal variation in adult height ('narrow-sense heritability') is approximately 80% (refs. 1–3). Previous analysis of genome-wide association studies (GWAS) of adult height showed that common variants together account for 50% of this heritable contribution to variation in height[4,5]. The most recent GWAS of adult height identified 180 loci, which together highlighted many genes relevant to human skeletal growth that had not been implicated in previous studies[6]. Common variants in these loci, however, only accounted for 10% of the phenotypic variation (~12% of heritability). Here we report the results from a GWAS meta-analysis of adult height in 253,288 individuals of European ancestry. We show that the additive contributions of fewer than 10,000 SNPs (at $P < 5 \times 10^{-3}$) can account for 36% of the heritability of adult height. Variants reaching genome-wide significance ($P < 5 \times 10^{-8}$) in this larger study (697 SNPs) clustered in loci, were substantially enriched for regulatory variants and implicated multiple known and previously unknown genes and pathways relevant to growth. More broadly, our results provide evidence that increasing GWAS sample sizes to the order of $10^5$ individuals, now plausible for many common traits, will likely continue to identifying associated variants and loci while improving knowledge of the biology of these traits.

## RESULTS

The overall analysis strategy is illustrated in **Supplementary Figure 1**. We first performed a GWAS meta-analysis of adult height using the summary statistics from 79 studies consisting of 253,288 individuals of European ancestry (Online Methods). We identified 697 SNPs that reached genome-wide significance ($P < 5 \times 10^{-8}$) using an approximate conditional and joint multiple-SNP (COJO) analysis[7] in GCTA[8] (Online Methods), which takes linkage disequilibrium (LD) between SNPs into account (**Supplementary Figs. 2** and **3**, and **Supplementary Table 1**). The 697 SNPs clustered in 423 loci, with a locus defined as one or multiple jointly associated SNPs located within a 1-Mb window. Most of these 697 SNPs were uncorrelated, although those in close physical proximity (for example, <1 Mb apart) might be in partial LD (see **Supplementary Table 1** for LD between adjacent pairs of the 697 SNPs). The clustering of signals was non-random (empirical enrichment of 1.4-fold; $P < 1 \times 10^{-4}$) with 90, 26 and 31 loci containing 2, 3 and ≥4 signals, respectively (**Supplementary Tables 1** and **2**, and **Supplementary Note**). We observed strong evidence of clustering of association signals within loci across a range of locus sizes, from 100 kb to 1.25 Mb, but the clustering was almost entirely driven by variants within 250 kb of index SNPs (**Supplementary Table 2** and **Supplementary Note**). In some loci, multiple signals clustered tightly around a single gene, whereas in other cases the clustering of associated variants was likely due to multiple different height-related genes being in close proximity (**Fig. 1** and **Supplementary Fig. 4**).

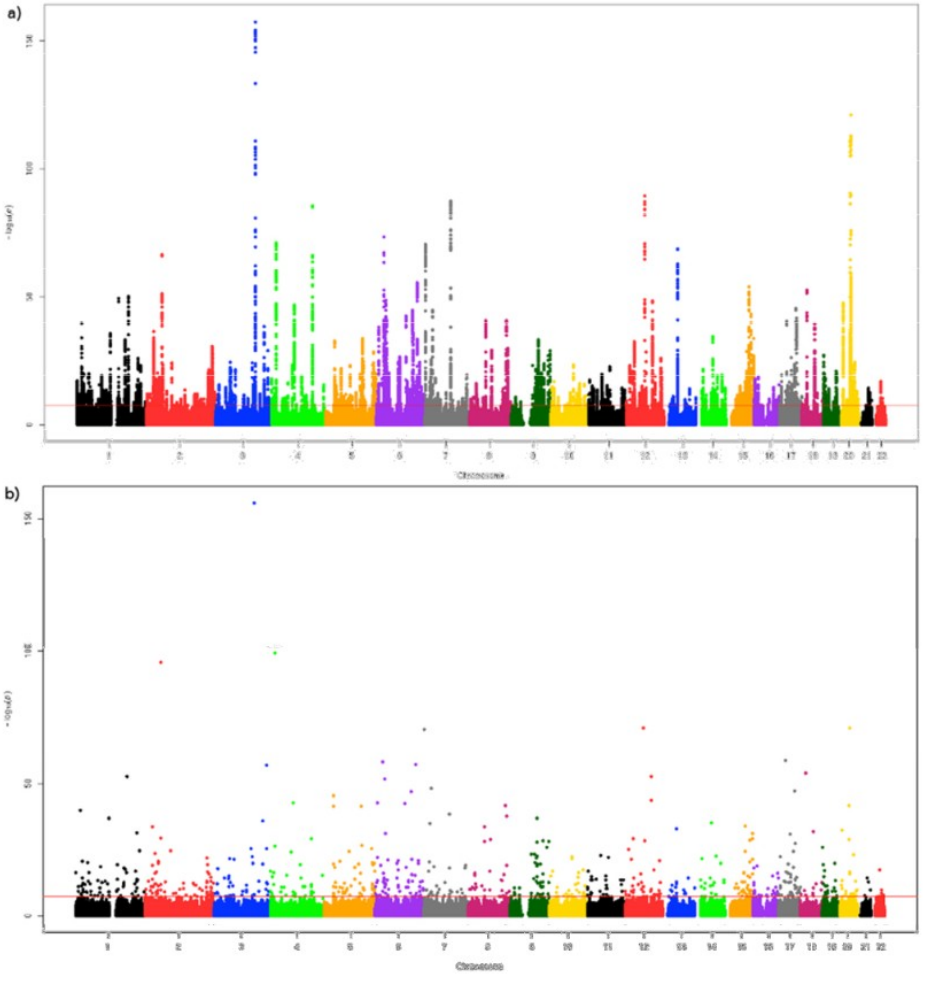
Of the 697 SNPs, 403 were represented on the Metabochip array[9]. Using data from 80,067 individuals genotyped on the Metabochip array from 37 independent studies, we observed very strong evidence of concordance of effect sizes between the Metabochip and GWAS samples ($P = 1.9 \times 10^{-160}$); >99% of variants had directionally consistent effects in the Metabochip and GWAS data (Online Methods, **Supplementary Table 3** and **Supplementary Note**).

We observed a large genome-wide 'inflation' of the test statistic for association, even after we corrected each study's test statistics by its individual inflation factor (single $\lambda_{GC} = 1.94$). At least two phenomena could have contributed to this observation. First, as described previously[10], highly polygenic models of inheritance are expected to increase the genomic inflation factor to levels comparable to what we observe. Second, height is particularly susceptible to confounding by population ancestry (stratification), which can also lead to inflation

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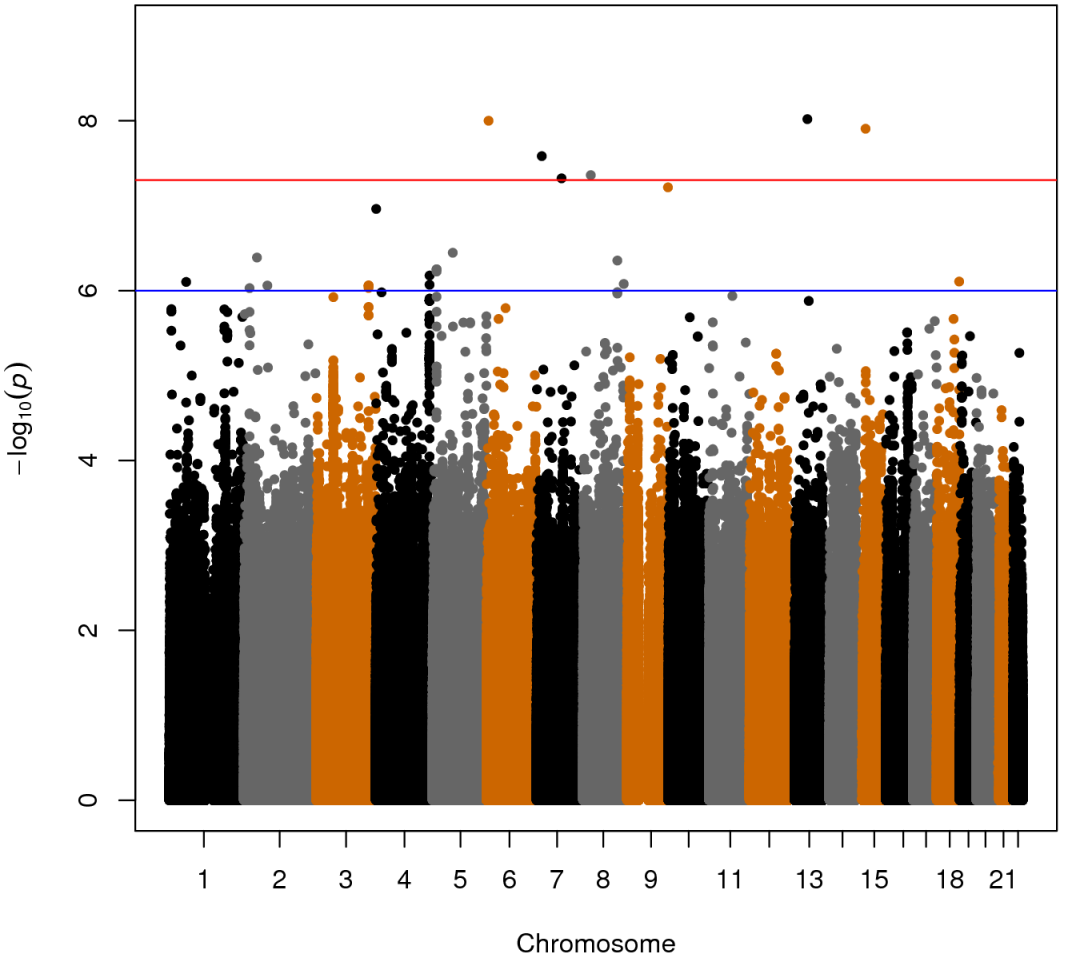


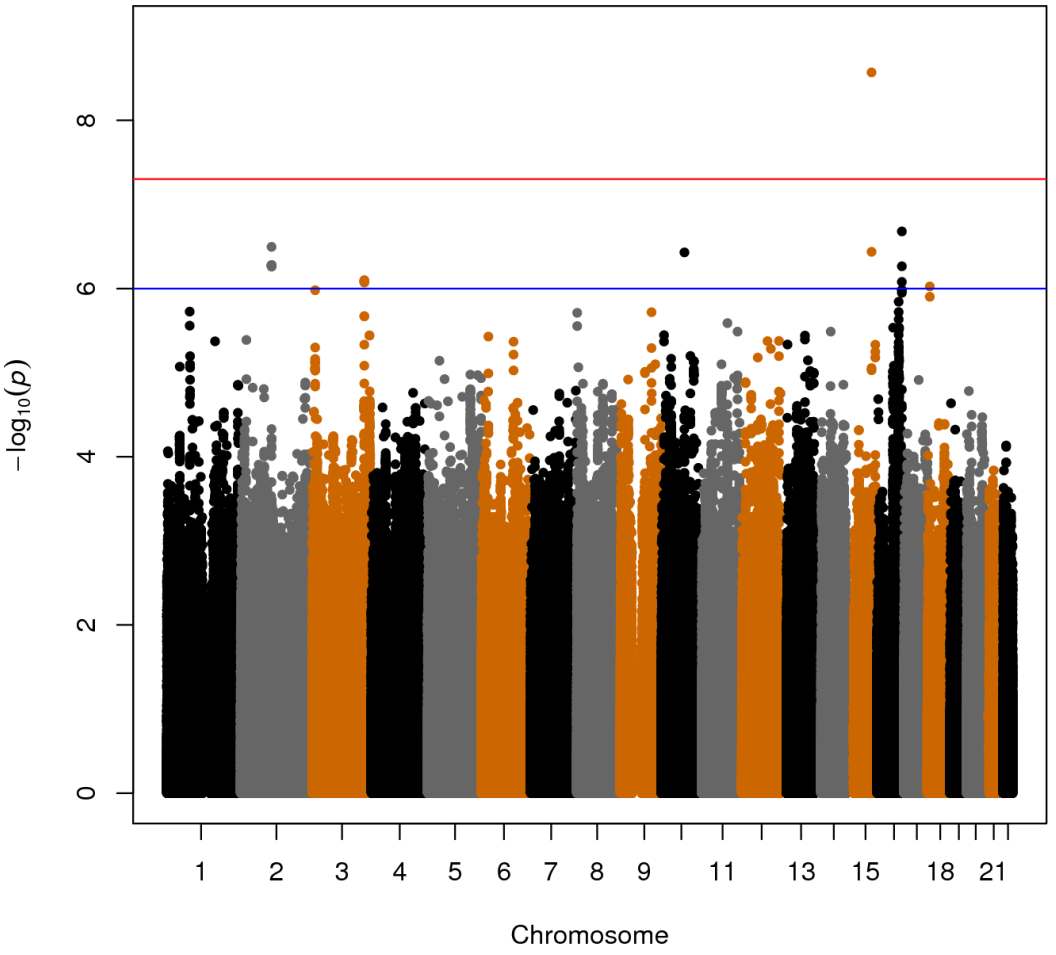

**Supplementary Figure 3**

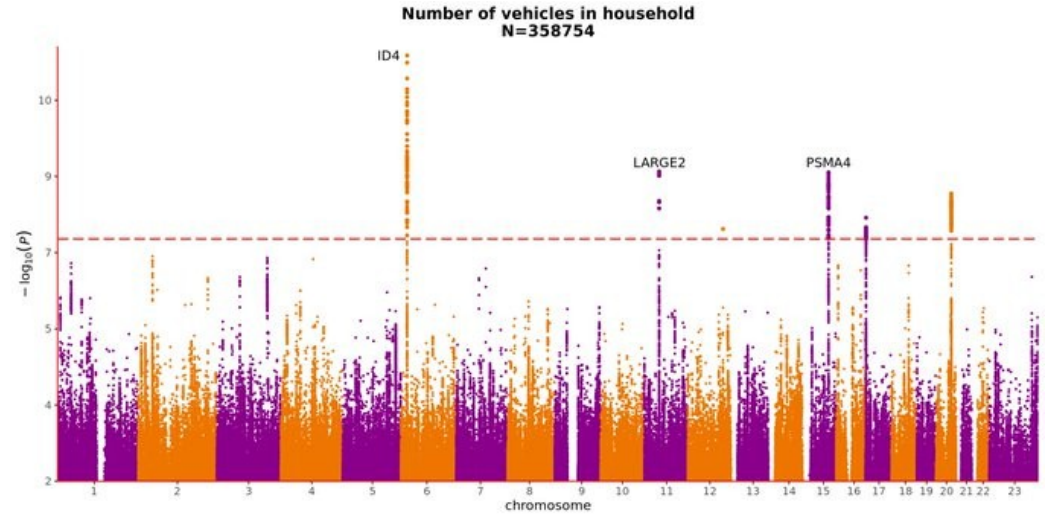
**Manhattan plots.**

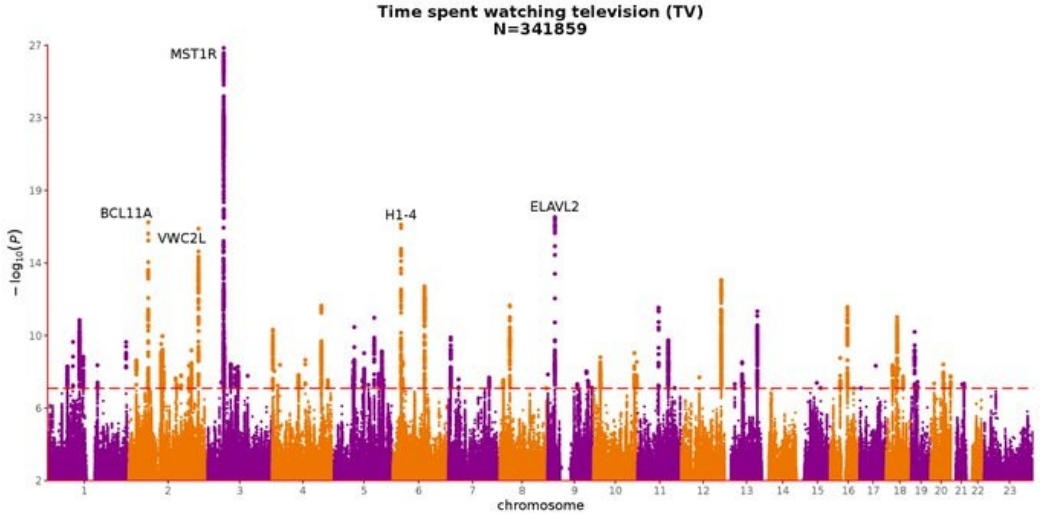
Plots of (**a**) results obtained from meta-analysis after applying a single genomic control correction and (**b**) results obtained from performing an approximate conditional and joint multiple-SNP analysis on the meta-analysis results. The red horizontal line in both plots represents the genome-wide significance threshold of $P = 5 \times 10^{-8}$.

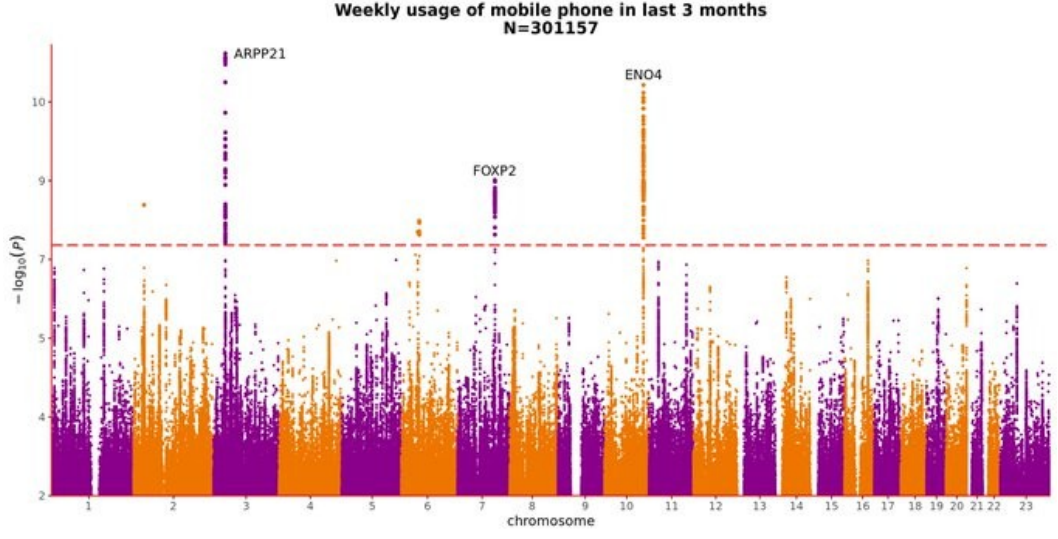
# Height genome-wide association Manhattan plots (~3000 individuals)

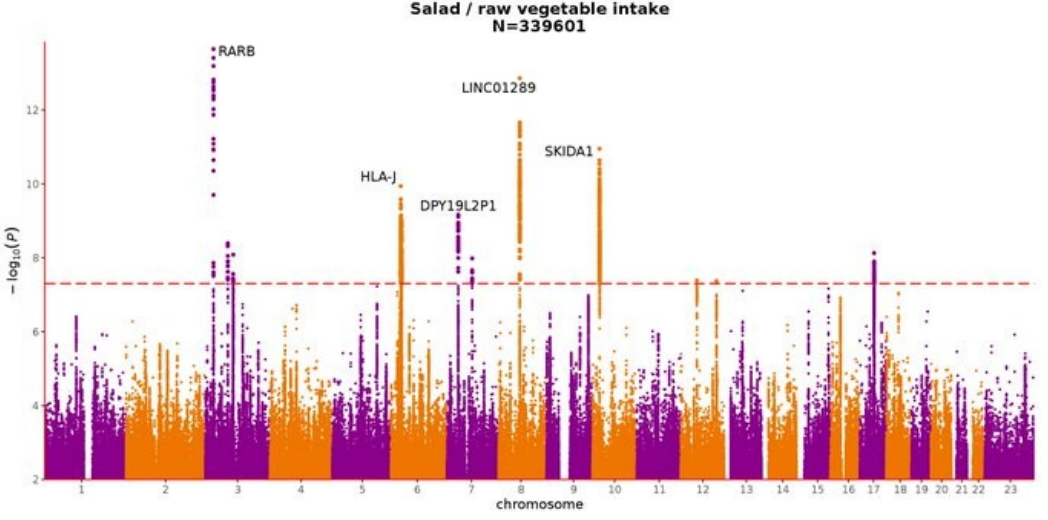


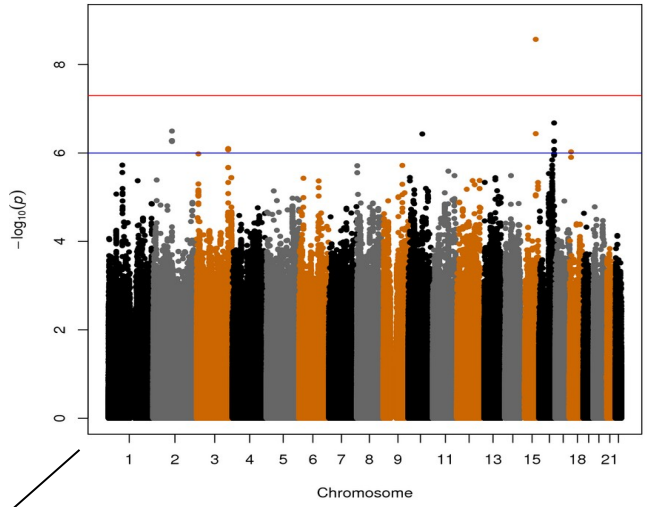

Number of vehicles in household
N=358754
ID4
LARGE2
PSMA4
$-\log_{10}(P)$
chromosome


Time spent watching television (TV)
N=341859
MST1R
BCL11A
VWC2L
H1-4
ELAVL2
$-\log_{10}(P)$
chromosome


Weekly usage of mobile phone in last 3 months
N=301157
ARPP21
FOXP2
ENO4
$-\log_{10}(P)$
chromosome


Salad / raw vegetable intake
N=339601
RARB
LINC01289
SKIDA1
HLA-J
DPY19L2P1
$-\log_{10}(P)$
chromosome

## DISCOVERY COHORT:




IDs of SNPs associated with trait:
SNP1, SNP2 ...

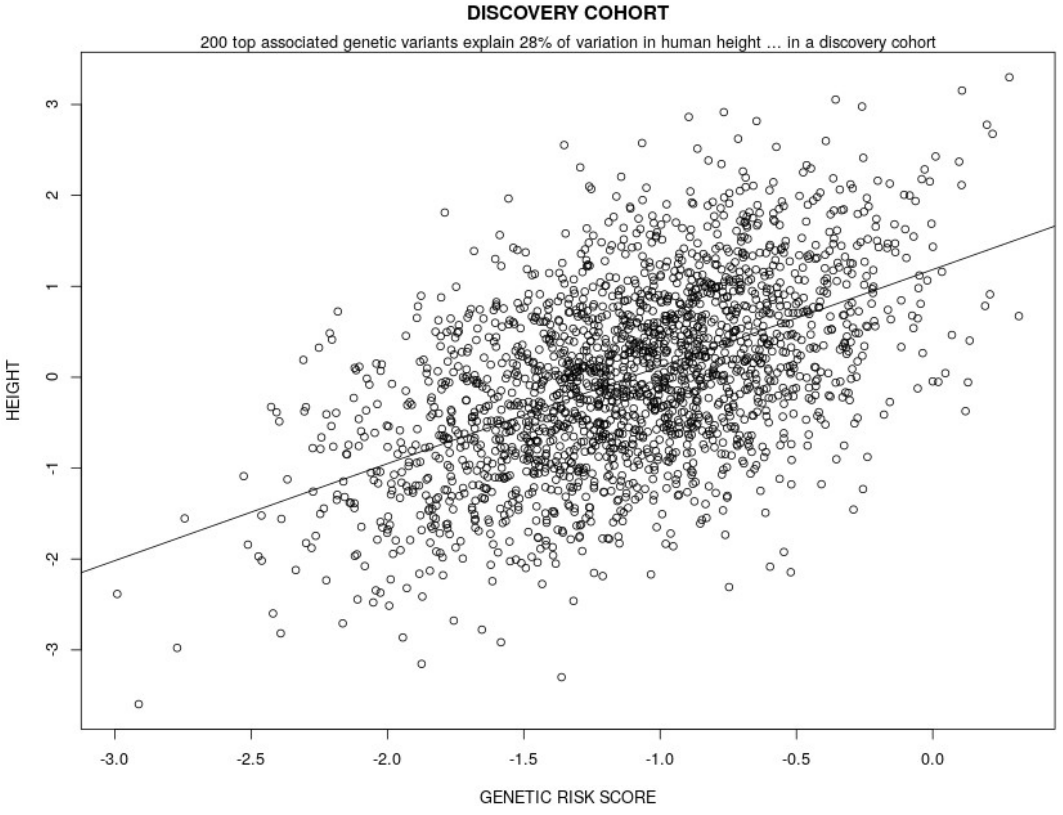
Beta-effect-sizes for those SNPs:
beta1, beta2 ...

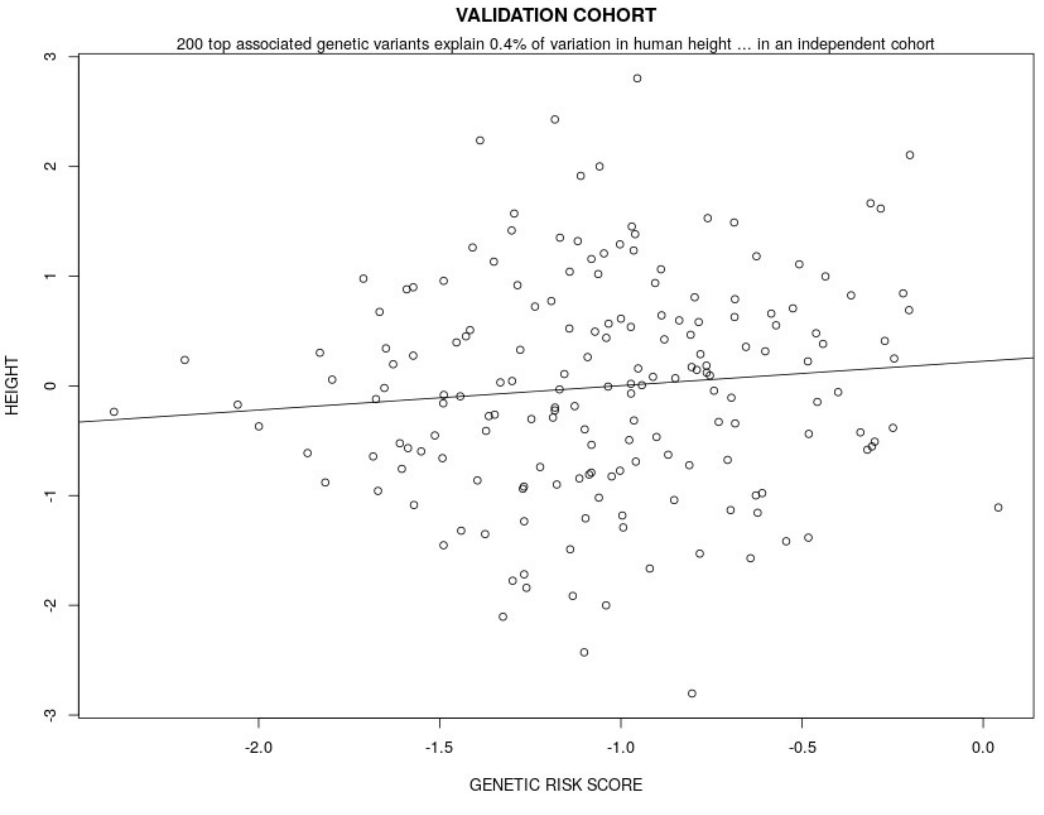
## VALIDATION COHORT:

Actual genotypes for SNP1, SNP2 ...
from the validation cohort

GRS = beta1 * SNP1 + beta2 * SNP2 + … ← training

Height_validation ~ GRS => Adjusted R^2 ← validation




GRS = $beta1_{discovery}$ * $SNP1_{discovery}$ + $beta2_{discovery}$ * $SNP2_{discovery}$ + …

Height ~ GRS => Adjusted R^2

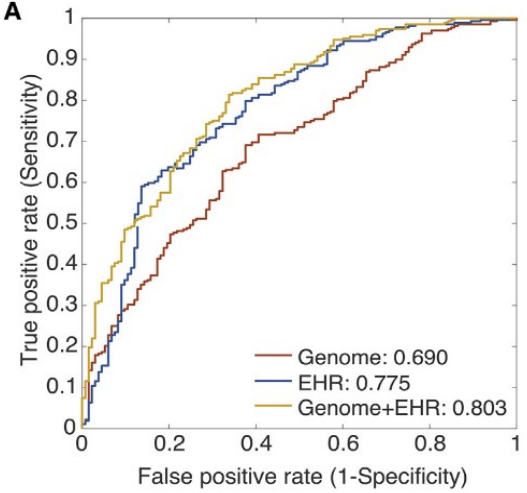
GRS = $beta1_{discovery}$ * $SNP1_{validation}$ + $beta2_{discovery}$ * $SNP2_{validation}$ + …

Height ~ GRS => Adjusted R^2

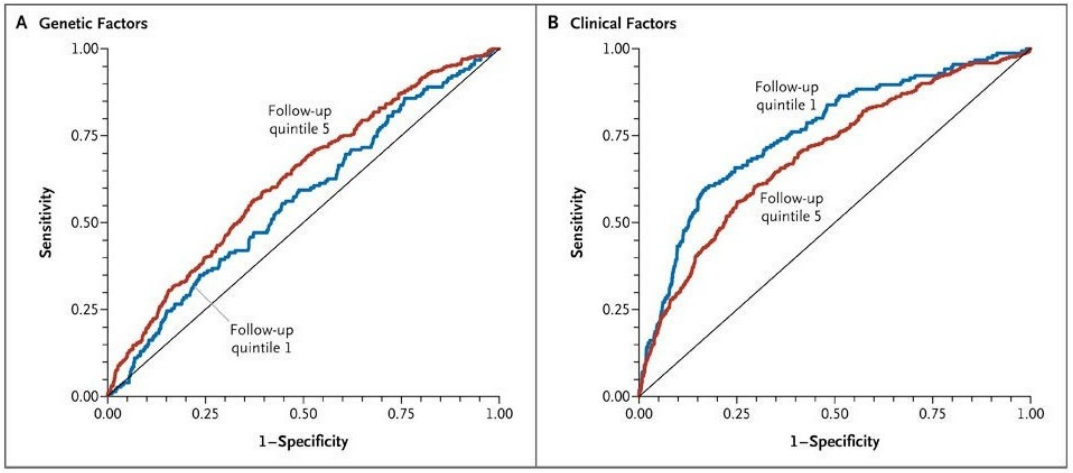


# The case of the missing heritability

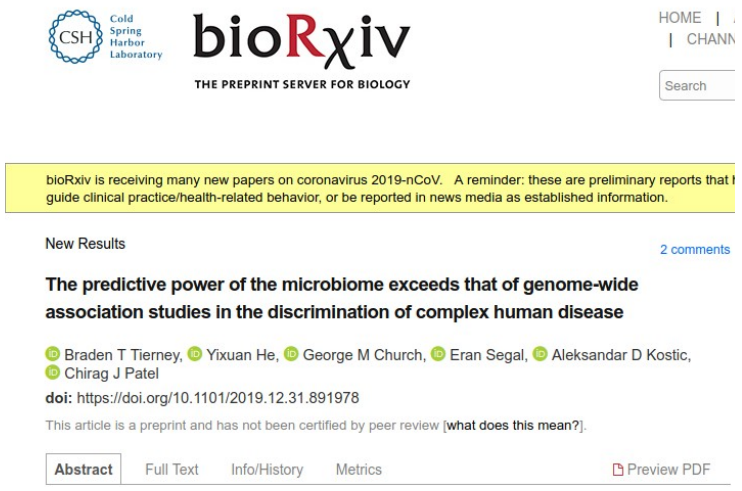
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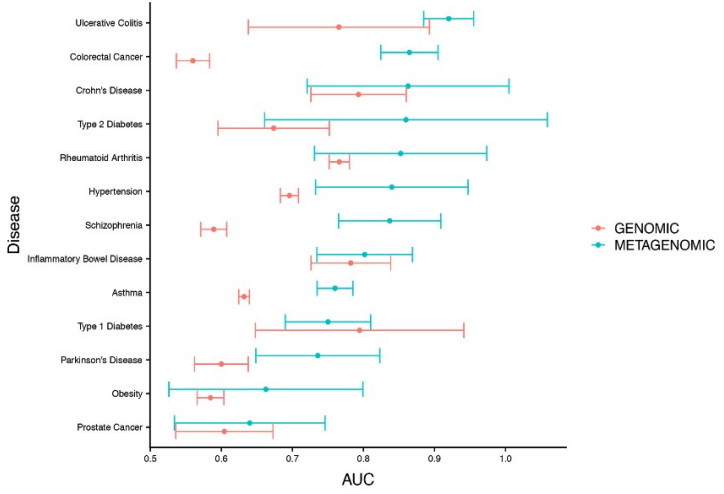



Li et al., 2018, Cell 174, 1361-1372
Decoding the Genomics of Abdominal Aortic Aneurism




Lyssenko et al., 2008, New England Journal of Medicine 359, 2220-2232
Clinical Risk Factors, DNA Variants and Development of Type 2 Diabetes



# Human Height Missing Heritability Case Solved



NEWS · 23 APRIL 2019

## Genetic study homes in on height's heritability mystery

Analysis of 20,000 genomes reveals that rare gene variants can help to explain how the trait is inherited.

Linda Geddes



Height is often inherited — but the precise genetics factors have long eluded researchers. Credit: Shomos Uddin/Getty

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### Recovery of trait heritability from whole genome sequence data

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**Abstract**

Heritability, the proportion of phenotypic variance explained by genetic factors, can be estimated from pedigree data [1], but such estimates are uninformative with respect to the underlying genetic architecture. Analyses of data from genome-wide association studies (GWAS) on unrelated individuals have shown that for human traits and disease, approximately one-third to two-thirds of heritability is captured by common SNPs [2–5]. It is not known whether the remaining heritability is due to the imperfect tagging of causal variants by common SNPs, in particular if the causal variants are rare, or other reasons such as over-estimation of heritability from pedigree data. Here we show that pedigree heritability for height and body mass index (BMI) appears to be fully recovered from whole-genome sequence (WGS) data on 21,620 unrelated individuals of European ancestry. We assigned 47.1 million genetic variants

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